

UI Redressing Mayhem: HttpOnly bypass PayPwn style

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Content

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In the previous post, a new cross-domain extraction method - affecting the latest version of the Mozilla Firefox browser - has been presented. The *iframe-to-iframe technique* was successfully used in a UI Redressing attack affecting LinkedIn. Today, I'm introducing an instance of the aforementioned method that involves a known Apache Web Server security issue, in order to steal session cookies that are protected by [HttpOnly](#) flag, thus allowing the attacker to perform [Session Hijacking](#) attacks. A new attack targeting PayPal systems will be also presented.

CVE-2012-0053: HttpOnly bypass and beyond

In January 2012 - even if the Apache defect was [known](#) and exploited for a while - [Norman Hippert](#) disclosed [CVE-2012-0053](#) bug affecting the Apache Web Server. The software was not able to correctly restrict an [header field](#) information disclosure in case of overlong or malformed HTTP requests. The vulnerability could be effectively combined with a [Cross-Site Scripting](#) attack to bypass the protection mechanism introduced by the [HttpOnly](#) flag and steal any session token stored as cookies value. Infact, an XSS vector could manipulate the [document.cookie](#) object to set an overlong cookie field, and forward a malformed request to the affected Apache Web Server with the intention to trigger the error message and *extract* the desiderated session cookies. The Apache bug can be abused in a series of attack scenarios such as the following:

- Bypassing [HttpOnly](#) flag with a XSS vulnerability on the same domain that is affected by the CVE-2012-0053;
- Bypassing the limitation introduced by cookie [path](#) whereas the XSS vulnerability affects a web resources that resides *outside* the defined path itself;

- Bypassing HttpOnly flag if a XSS vulnerability is found on *any* subdomains of the host that is affected by the Apache disclosure issue, if exploited in conjunction with a UI Redressing attack - that allows the cross-domain content extraction of the information included in the triggered Apache error message.

It should also be noted that the Apache Web Server is often used as a *reverse proxy* configuration. As a result, any session object on any server-side technology, could be attacked with the described vectors.

Smashing PayPal for Fun but.. NO Profit

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During my security research on UI Redressing attacks I found multiple PayPal subdomains (e.g. **https://b.stats.paypal.com**) affected by the Apache disclosure bug as detailed in Figure 1 and Figure 2.

|  (https://web.archive.org/web/20170622123259/http://4.bp.blogspot.com/-UX7AXFvyOAw/UM0VW0KNetI/AAAAAAAAAFA/F6G9tYypnHs/s1600/resp.png) || Figure 1 - HTTP request with the overlong cookie. ||

|  (https://web.archive.org/web/20170622123259/http://1.bp.blogspot.com/-GR6r3FjU_X8/UM0VVcZ2DRI/AAAAAAAAAE4/aPydP9Sii60/s1600/req.png) || Figure 2 - Apache error message with the disclosure of the malformed Cookie header. ||

Despite in the first instance the bug could appear as useless, I found that the PayPal application - **www.paypal.com** - delivers the session cookies defining the domain to **.paypal.com** (Figure 3 and Figure 4).

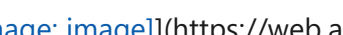
|  (https://web.archive.org/web/20170622123259/http://3.bp.blogspot.com/-dsEPEISXMEg/UNDgfkQbkwI/AAAAAAAAAGI/3X3dIsZkBUM/s1600/sess_1.png) || Figure 3 - Post-authentication cookies delivery. ||

|  (https://web.archive.org/web/20170622123259/http://1.bp.blogspot.com/-bO49I3nAMX8/UNDgpy1wEyl/AAAAAAAAAGQ/ZoUJUptfedk/s1600/sess_2.png) || Figure 4 - Cookies delivered to the personal.paypal.com subdomain. ||

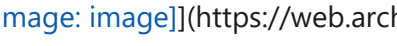
The highlighted security issues could be abused to attack authenticated PayPal users, implementing the mentioned UI Redressing attacks combined with the cookie disclosure bug. But.. I had a problem: I had **no** XSS vulnerability on any PayPal web application - not that there're not! I was able to circumnavigate the limitation identifying another vulnerability on a different PayPal subdomain, that allowed me to define a *monster cookie* with a single HTTP request. As first, please note the following URL:

- **https://history.paypal.com/helpcenter/script/pphc_rating.js.jsp?locale=&_dyncharset=UTF-8&countrycode=CA&cmd=_AAAAKKKKKKAAAAKKKKKK[.very long string here...]KKKKKKAAAAKKKKKKAAAAKKKKKK&serverInstance=9002&no_strip= **

As detailed in Figure 5, the navigation of the above URL involves the setting of the cookie named **navcmd** *and then a bit of client-side black magic* defines two new cookie fields named ****s_sess** and **s_pers** (Figure 6) that complete the desiderated malformed HTTP request.

|  (https://web.archive.org/web/20170622123259/http://4.bp.blogspot.com/-F_KMDWywmt0/UNDpSV6sSal/AAAAAAAAAGg/wFCDDBBku1A/s1600/monster_1.png) || Figure 5 -

Cookie defined with attacker-controlled input. ||

|  (https://web.archive.org/web/20170622123259/http://4.bp.blogspot.com/-d2aNzoxHeBY/UNDr2TK8JOI/AAAAAAAAAGw/64DnokPaPLw/s1600/monster_2.png) || | Figure 6 - Final monster cookie. ||

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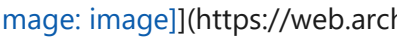
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The exploitation is now trivial. The following are the logical steps implemented by the Proof of Concept exploit:

- The exploit triggers the victim to open an *under pop* (Figure 7) web page that generates the monster cookie - with **domain=.paypal.com** - involving the **history.paypal.com** application;
- The **https://b.stats.paypal.com** is then framed thus inducing the forward of a malformed HTTP request that triggers the disclosure of the Cookie header, containing the PayPal account's session cookies;**

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- The malicious page allows the victim to play the d&d game with the extraction of the secret session cookies.

|  (https://web.archive.org/web/20170622123259/http://4.bp.blogspot.com/-JR4tHH-6l3o/UNDSIFyxqOI/AAAAAAAAAG8/euEk11tnemI/s1600/pop_under.png) || | Figure 7 - Pop-under page with the navigation of the monster cookie's generation URL. ||

The attacker now holds the cookies that can be used to perform a Session Hijacking attack against the victim's PayPal account. A working Proof of Concept has been developed and can be download [here](#). The following is a video that illustrates the described attack:

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